
Learning Grid Cells with World Value Functions

Sergio A. Frasco*

School of Computer Science and Applied Mathematics
University of the Witwatersrand
Johannesburg, South Africa
2427724@students.wits.ac.za

Clémentine Dominé

Gatsby Computational Neuroscience Unit
University College London
London, United Kingdom

Devon Jarvis, Geraud N. Tasse

School of Computer Science and Applied Mathematics
University of the Witwatersrand
Johannesburg, South Africa

Abstract

Spatial navigation in animals exhibits flexibility, often demonstrating *zero-shot* capabilities—the ability to perform tasks without prior experience, such as innate escape behaviours in mice [2]. Grid cells in the entorhinal cortex, which fire in distinctive hexagonal patterns, are crucial for navigation, especially in escape behaviour; however, their role in goal-oriented agents is unclear. Studies in mice show that goal-oriented learning alters grid cell activity and, while traditional Successor Representation (SR) models with random agents can generate hexagonal grid patterns [8], they typically learn the SRs from an established policy. Thus, behaviour precedes spatial representations. Here, we explore whether goal-oriented learning can support the simultaneous development of grid cells and a behaviour policy through World Value Functions (WVF) [5], offering another perspective on grid cell emergence. These WVFs are goal-oriented value functions allowing agents to solve not only specific tasks but also compositions of tasks within the training set [5]. Figure 3 demonstrates the main computation in our method. We compare the original SR baseline (SARSA and random policy producing the SR matrix) [8] to a novel WVF-based method which used Q-learning to obtain the SR matrix and simultaneously estimates the mean reward of each goal-oriented reward matrix. Crucially, we consider whether we can decompose WVFs into task-specific reward maps and a single, *shared* SR matrix learned during training. By taking the dot product between an estimated reward and learned SR matrix we are able to retrieve the goal-oriented value function. This joint learning process introduces tension, as the SR matrix must accommodate multiple goals, driving the emergence of grid cells. Unlike traditional approaches, where the SR is inferred after policy learning, our method integrates SR learning into the behaviour itself, allowing grid-like patterns to emerge as part of the goal-directed learning process.

Keywords: Reinforcement Learning, Grid Cells, World Value Functions

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1 Introduction

Reinforcement learning (RL) provides a powerful framework for understanding decision-making processes in both animals and artificial systems, particularly in the domain of spatial cognition [6]. Among the most fascinating aspects of spatial navigation is the remarkable adaptability displayed by animals, exemplified by mice’s ability to navigate novel environments and exhibit zero-shot navigation abilities, such as innate escape behaviours without prior learning [2]. This adaptive capability is fundamentally linked to specialized neural mechanisms, particularly the grid cells in the entorhinal cortex. Grid cells are a type of spatially selective neuron that exhibit distinctive hexagonal firing patterns during navigation. These patterns are thought to provide a coordinate system for spatial representations, with a population of grid cells (each with a distinct frequency and orientation) collectively encoding spatial locations across environments. While the discovery of grid cells has significantly advanced our understanding of spatial cognition, questions remain about how they are formed and contribute to goal-directed behaviour. The hexagonal structure of a grid cell can be seen in Figure 1(a) [7].

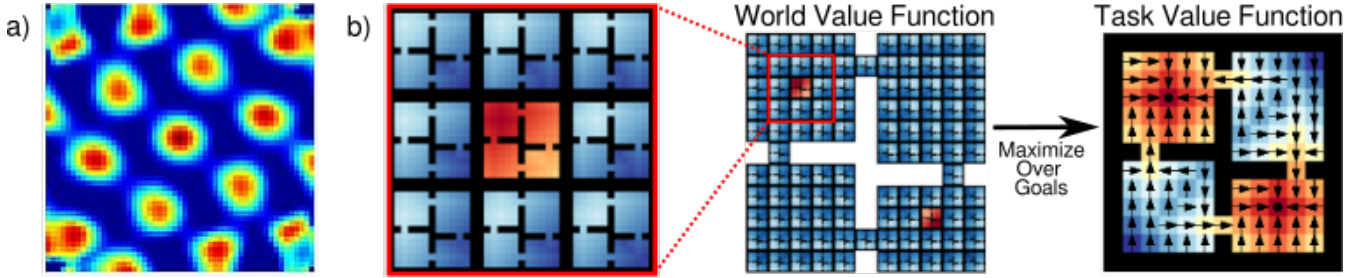


Figure 1: (a) Example grid cell representation from a single neuron in the entorhinal cortex [7]. (b) World value function representations of space in the 4-room problem. A single WVF (middle) represents the value of a particular state for every possible goal. In this example each state is also a goal and we zoom in on nine possible goal points (left). Note that the reward for reaching the goal point is still produced by the environment. Thus, the goal-oriented value functions are still relative to the real value of reaching the goal (hence only two goal points have high rewards and produce high values). By determining for each state, which goal is optimal to follow, we can produce a new value function which not only optimizes goal specific rewards, but also which goals are oriented towards (right).

Traditional approaches to modelling grid cells have often relied on the Successor Representations (SR) models where agents learn expected future states based on a given policy [8]. With this approach behaviour precedes spatial map formation as the agent must first learn a policy [3]. In the special case of a random policy a symmetric SR matrix is learned such that the eigendecomposition of the SR produces grid-like firing patterns, similar to the neuron depicted in Figure 1(a). However, the question remains: why would the representations to support navigation emerge after a policy is already learned. A consequence of this approach is that the grid cells produced by the SR model are skewed significantly by goals or salient points within the environment as it distorts the movement of the agent [8].

Recent developments in RL have introduced World Value Functions (WVFs). Unlike a traditional value function, which is tied to a specific reward function and policy, a WVF represents a set of value functions corresponding to all goals in the environment. An individual WVF is still associated with a specific reward function, but with penalties for achieving incorrect goals. By training a single WVF, an agent can compose it with any reward function to capture complex, goal-driven behaviours zero-shot [9] (see Figure 1(b)).

In this work, we consider whether a combination of WVFs and SRs can produce a model of grid cell formation and behaviour learning, where both are learned concurrently. Specifically, we consider sharing a single SR matrix across all goal-directed value functions within the WVFs. We find that this introduces a strong regularization effect as the SR matrix must adjust to accommodate multiple goal points within an environment. By leveraging this shared SR structure, we hypothesize that agents can achieve a more balanced spatial representation and develop grid cell patterns that align more naturally with goal-directed behaviour. As a first step towards determining the accuracy of our hypothesis, here we consider a single-room domain, a standard task for studying spatial navigation with SRs, and compare to the established regular-valued (SARSA) baseline SR.

2 Background

The SR provides a framework for predictive state representations in reinforcement learning [3]. Instead of estimating future rewards of a given policy directly, SR predicts the future state occupancies, allowing value computation by combining these predictions with a reward model. This factorization separates the environment’s dynamics under a given policy from task-specific rewards, enabling the agent to reuse the SR across tasks by updating only the reward component. Hence, building adaptable world models which promote transferability.

Once the SR of a given policy is learned, any new value function for that policy can be computed by combining the SR with the task’s reward predictions [3]. Gershman [4] further explores SR’s theoretical basis, asserting it enhances flexibility and transfer in reinforcement learning. Mathematically, the SR, denoted as $\Phi^\pi(s, g)$, captures the discounted future occupancy of state g from state s under policy π . Specifically, $\Phi^\pi \in \mathbb{R}^{|S| \times |S|}$, where $|S|$ is the number of discrete states in the environment. Each entry in the matrix represents the expected discounted frequency with which the agent will occupy state g when starting from state s and following the policy π , providing a compact representation of state transitions over time:

$$\Phi^\pi(s_t, g_t) = \Phi^\pi(s_t, g_t) + \alpha(1[s_t = g_t] + \gamma(\Phi^\pi(s_{t+1}, g_t) - \Phi^\pi(s_t, g_t))),$$

where γ is the discount factor. The value function $V^\pi(s)$ is then given by:

$$V^\pi(s) = R(s) \cdot \Phi^\pi(s, \cdot)$$

where $R(s)$ represents the expected reward at state s . Before flattening, R is a three-dimensional map of expected rewards corresponding to each point in the grid, where each *slice* of this matrix represents a unique state with a goal present. For the matrix multiplication to be valid, $R(s)$ is flattened into a 2D plane, allowing it to align with the dimensions of the matrix $\Phi^\pi(s, g)$. This factorization allows for reusing $\Phi^\pi(s, g)$ across tasks while updating the reward map. Let R_{new} represent a new reward map, then the corresponding value function can be expressed as $V_{new}^\pi(s) = R_{new}(s) \cdot \Phi^\pi(s, \cdot)$. This notation emphasizes that while $\Phi^\pi(s, g)$ remains constant, R_{new} can be updated to reflect changes in the rewards for different tasks.

Following on from this, Predictive Mapping Theory [8] suggests that the hippocampus encodes an environmental map linking states and rewards. This map supports navigation by allowing evaluation of trajectories and task generalization. Similarly, grid cells in the entorhinal cortex exhibit grid-like firing patterns for spatial representation. Banino [1] found grid-like patterns in agent representations trained with deep reinforcement learning, suggesting that SR may underpin predictive mapping and generalization in the brain’s spatial and memory circuits. Moreover, SR’s grid-like structure, as seen through its eigendecomposition, highlights its potential as a computational mechanism for grid cells in spatial navigation [8].

2.1 World Value Functions

World Value Functions are a novel approach in reinforcement learning that enhance an agent’s ability to learn and plan by encoding how to achieve various goals within its environment [9]. WVF’s are a type of goal-oriented general value function $Q(s, g, a)$ that enables agents not only to maximise specific task rewards $R(s, a, s')$ but also to tackle any goal-reaching task they encounter. This is achieved by defining an internal goal space $\mathcal{G}$, which consists of all states where the agent experiences a terminal transition. A corresponding internal reward function is then defined to penalise the agent (with a penalty R_{MIN}) for achieving the wrong goals:

$$R(s, g, a, s') = R(s, a, s') \text{ if } (s \neq g \text{ and } s' \text{ is terminal}) \text{ else } R_{MIN}.$$

To obtain the policy $\pi^*(s)$ to solve a given task after learning a WVF $Q^*(s, g, a)$, the agent simply maximises the WVF over goals and actions: $\pi^*(s) = \operatorname{argmax}_a \max_g Q^*(s, g, a)$.

Nangue Tasse [9] demonstrates that WVF’s can be learned from a single stream of experience without requiring additional information or modifications to the standard reinforcement learning framework. This characteristic allows agents to compute optimal policies for new tasks by estimating the task’s reward function based on the learned WVF. Experimental results suggest that while agents may experience a longer initial learning phase with WVF’s compared to traditional value functions, the potential for accelerated learning in multi-task scenarios becomes evident once sufficient initial WVF’s are established. The composition of learned WVF’s enables agents to adapt more quickly to new tasks, provided that the dynamics of the environment remain consistent. However, it is crucial to note that WVF’s do not infer the environment’s dynamics; rather, they rely on the assumption that the environment’s dynamics do not change. If dynamics shift, the composition of previously learned policies may not effectively support new tasks, which highlights an important assumption in our approach to sharing the SR matrix.

3 Training World Value Functions and Successor Representations: Methodology and Results

To train the WVF and develop our shared SR, we decompose the WVF into a reward matrix and an SR matrix as seen in Figure 2:

1. Reward Matrix

- Each slice in the rewards matrix represents the mean rewards for each state, corresponding to specific goal-oriented tasks. These slices are presented as row-vectors during training, ensuring balanced training to all goal states. Training cycles through these reward slices uniformly to prevent over-fitting to specific goals.

2. Shared SR Matrix:

- Each row in the SR matrix captures the discounted probability of reaching each state under a goal-oriented policy. This SR matrix reflects an agent’s expectations about future state transitions given the best policy (highlighted in blue in the equation below). This matrix is updated using Q-learning by utilizing the maximum SR value for the next state. That is, where $\rho(s, a)$ is the transition function:

$$\Phi^*(s_t, g_t) = \Phi^*(s_t, g_t) + \alpha(1[s_t = g_t] + \gamma(\max_a \Phi^*(\rho(s, a), g_t) - \Phi^*(s_t, g_t))),$$

- This process contrasts the baseline method, where the SR matrix is updated via SARSA by utilizing the SR value for the next state under a random policy π (highlighted in red in the equation below) [8]:

$$\Phi^\pi(s_t, g_t) = \Phi^\pi(s_t, g_t) + \alpha(1[s_t = g_t] + \gamma(\Phi^\pi(\rho(s, \pi(s)), g_t) - \Phi^\pi(s_t, g_t))).$$

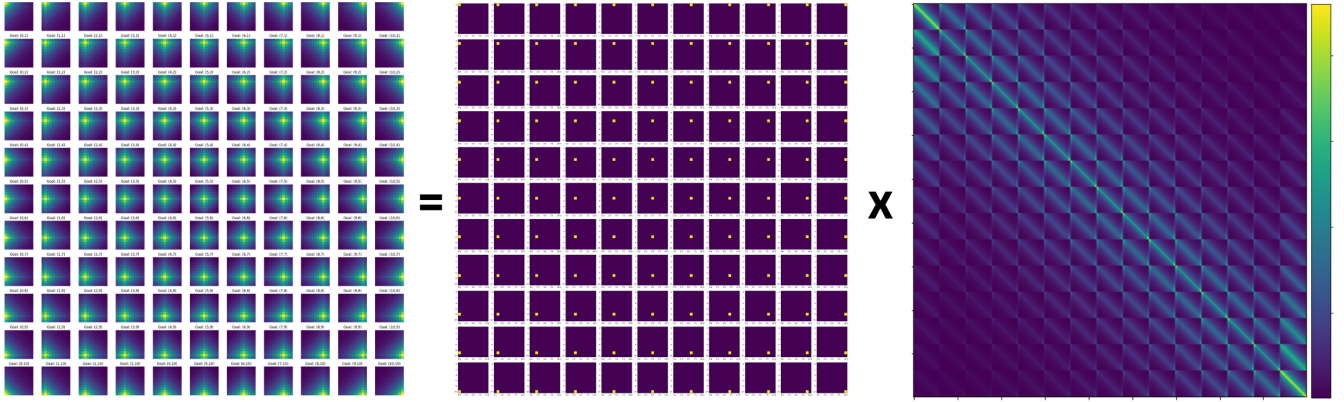


Figure 2: **Decomposition of the WVF into a set of goal-oriented reward matrices and a shared SR matrix:** The WVF (left) is shown as the result of the dot product between distinct reward structures (middle) and a shared SR matrix (right). Each reward slice (a square map in the middle matrix) corresponds to a specific goal, while the shared SR matrix captures the underlying state transitions, independent of task. To obtain the value function a reward slice is flattened and dot product with the SR matrix. We hypothesise that the sharing of the SR matrix acts as a strong regulariser towards more realistic grid cell formations.

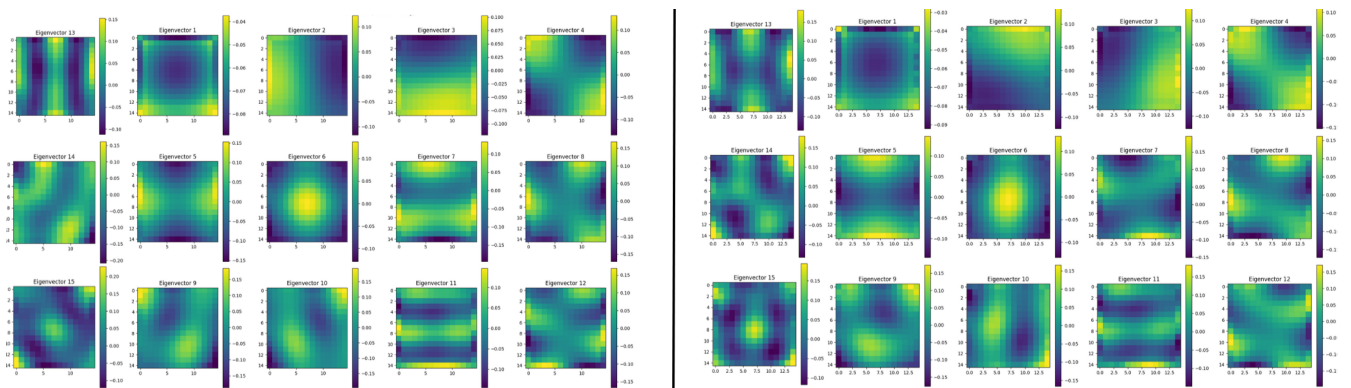


Figure 3: **WVF SRs recover grid cells similar to the ideal SR matrix with a random policy:** (left) Illustration of the grid cells that emerge from the eigendecomposition of the SR matrix trained using SARSA with a random policy. (right) Illustration of the grid cells that emerge from the eigendecomposition of the shared SR matrix trained using Q-learning on Goal-oriented policies and combined into a WVF.

Figure 3 depicts the eigenvectors of the shared SR matrix and shows that the decomposed WVF produces grid-like patterns comparable to the traditional SR model [8]. Thus, goal-oriented learning through WVFs can yield grid-like representations similar to those produced by traditional random agents, which have been shown to resemble biological grid cells. The emergence of these grid-like patterns in our WVF-based model hints at a potential mechanism by which animals might develop and utilize spatial representations in a goal-directed manner. This finding aligns with the observed zero-shot capabilities in animals, such as innate escape behaviours in mice [2], as WVFs have been shown to support zero-shot generalization to new tasks [5]. However, our current focus has been solely on a dense reward space, but it is seldom the case that all points in a room are salient. Thus, we explore varying densities of goal points to assess their effects on grid cell behaviour, specifically through a metric that quantifies grid cell formation: the gridness score. This score is calculated by analyzing the firing rate of an agent’s neurons in various states in a grid environment, measuring the degree of regularity or hexagonal patterning these form.

This experiment demonstrates that goal-oriented learning through WVF can create grid-like representations similar to traditional methods, connecting goal-oriented behaviour with grid cell emergence. While SR depends on the agent’s policy, WVF allows adaptability to different reward structures by multiplying the SR matrix with various reward vectors, facilitating task composition across diverse environments. These findings connect computational models with animal behaviour and deepen our understanding of spatial cognition, highlighting how learning dense transition points supports task generalization, a critical feature of WVF.

4 Discussion

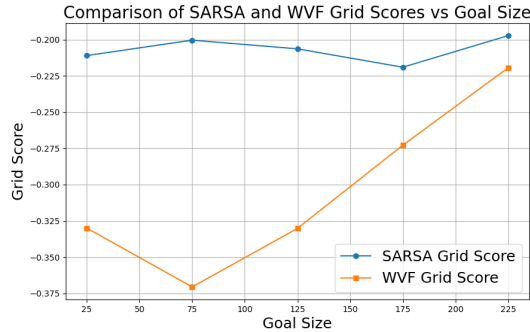


Figure 4: Impact of Goal Density on Grid Scores: This figure compares the grid scores of the baseline SARSA approach and our novel WVF method as the number of goal states increases. Higher scores indicate stronger grid-like representations. While SARSA maintains consistent grid scores, WVF improves as goal density increases, matching SARSA when goals are in every state. This suggests WVF effectively generates grid-like patterns, with higher scores indicating more efficient navigation and animal-like spatial representations.

This research has broad implications beyond theoretical neuroscience. Understanding how the brain navigates space could offer insights into learning and decision-making for artificial systems, as demonstrated by applications inspired by entorhinal grid cells in neural networks for navigation and spatial memory [1]. Additionally, these findings could impact robotics and AI, helping develop intelligent agents capable of flexible navigation and planning. Since forming maps is fundamental to human cognition, with the hippocampus playing a key role in memory, this suggests humans create conceptual maps that link memories to spatial contexts. These ideas could inspire new approaches to memory and reasoning in artificial systems.

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